

KAROL ROSZAK

Generative AI Engineer | Agentic Systems | Diffusion & LLM Pipelines | GCP Infrastructure

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years designing and shipping production AI systems end to end for clients across US, UK and EU. Built agentic workflows, multi-modal generation pipelines (text, image, video, voice) and production deployment serving on GCP. As Founding Lead AI at a US stealth startup, architected a multi-modal platform that powered an internal e-commerce engine generating advertising assets across thousands of SKUs. Deep expertise in GenAI, agentic systems and Claude-Code-driven engineering, Python, Docker and GCP-based infrastructure. Led international engineering teams and advised C-suite on AI strategy.

TECHNICAL SKILLS

Core. Python ecosystem (FastAPI, PyTorch, CrewAI, Diffusers), Docker, GCP, Runpod, CI/CD, REST APIs

AI Development Tools. Claude Code, Codex, Opencode, spec-driven development, sub-agent orchestration

AI Engineering. LLM application development, RAG, vector search, prompt & context engineering, agentic workflows (CrewAI, LangChain, LangGraph), vLLM, AI safety

Generative AI (Image / Video). Image and video generation pipelines (ComfyUI), diffusion model fine-tuning & serving, LoRA training (Kohya-ss), ControlNets, IP-Adapters, identity preservation, virtual try-on, lip-sync

Data & Storage. PostgreSQL, ChromaDB, Pinecone

LLMs. GPT, Claude, Gemini, Llama, Mistral, DeepSeek, Qwen

Miscellaneous. Runpod, Replicate, Vertex AI, SDXL, Flux, Qwen Image, WAN, LTX, Nano Banana, GPT Image, Kling, Seedance 2.0, Higgsfield, Midjourney

PROFESSIONAL EXPERIENCE

Generative AI Engineer & Consultant / Independent

Sep 2024 - Present / Remote

Sole proprietorship: end-to-end GenAI delivery (agentic automation, multi-modal pipelines, POC/MVP build-outs), AI consulting, enterprise integration, and team workshops for clients in EU and US.

Generative AI Engineer / takara.ai / Contract

Jul - Sep 2025 / London, UK (Remote)

- Achieved >99% identity and style consistency at production scale by fine-tuning SOTA diffusion models and training hundreds of LoRAs
- Engineered ComfyUI pipelines for editing, style transfer and identity preservation that cut graphic-designer asset turnaround from days to hours
- Extended workflow from image to video, delivering consistent-character clips that unlocked new client deliverables

Founding Lead AI Engineer / Stealth AI Startup / Contract

Jan - Jun 2025 / California, US (Remote)

- Replaced most of the manual graphic-design workflow at scale with autonomous ad-asset generation, enabling full campaigns with a fraction of the previous design headcount
- Delivered working POC in 2 weeks and production MVP in 8 weeks, powering automated ad-asset generation across a multi-thousand-product e-commerce catalog
- Designed GCP infrastructure that trained and served LoRAs for thousands of e-commerce SKUs, enabling on-demand ad creative generation
- Led international team of 5 engineers; aligned technical roadmap with business objectives and mentored the team

Generative AI Engineer / LABHOUSE / Full-time Sep 2024 - Feb 2025 / Barcelona, Spain (Remote)

- Powered a consumer face-generation app at scale: deployed Replicate serving stack behind iOS/Android clients, handling tens of thousands of generations/day at 99.9% uptime with auto-scaling
- Cut image generation pipeline processing time by 25%+, reducing compute costs across production workflows
- Built production ComfyUI workflows on Flux/SDXL for identity preservation, virtual try-on and lip-sync

Generative AI Engineer / indaHash / Contract Nov 2023 - Sep 2024 / Riyadh, Saudi Arabia (Remote)

- Eliminated manual content bottleneck with an LLM pipeline converting free-form briefs to structured input, cutting per-brief turnaround from hours to seconds
- Directed international AI-influencer program delivering ~1,000+ image and video assets per month across multiple personas at production quality
- Automated end-to-end image generation pipelines on SDXL with LoRA and ControlNet for high-resolution production output
- Mentored technical and non-technical staff on AI tool adoption, accelerating organization-wide AI integration

AI Engineer / Veibul / Contract Jan 2024 - Jul 2024 / Poznan, Poland (Remote)

- Led AI integration into flagship product with LLM and diffusion-based image generation pipelines, reducing deployment cost and inference latency
- Designed and secured RAG chatbots, AI agents and diffusion workflows with prompt-injection safeguards and hallucination checks
- Advised the CEO on emerging AI strategy and mentored developers in prompt engineering and deployment practices

Junior Data Scientist / Veibul / Contract Jan 2021 - Jan 2024 / Poznan, Poland (Hybrid)

- Built object-detection and segmentation systems (YOLO, SAM) for industrial use cases: strawberry-ripeness detection for automated picking, defect detection on cookie production lines, and fruit/vegetable sorting and counting on conveyor systems
- Designed, trained, and deployed AI models across on-premise and cloud environments (GCP, DigitalOcean, TogetherAI)
- Maintained labeled datasets and quality-control pipelines that measurably improved model accuracy on production CV models (defect detection, sorting, counting)

CERTIFICATIONS

Multi-Agent Systems with CrewAI / DeepLearning.AI Mar 2026

Google Cloud Certified Associate Cloud Engineer / Google Feb 2026

Leading People and Teams / University of Michigan Oct 2025

EDUCATION

Bachelor of Engineering (BE), Artificial Intelligence Oct 2019 - Feb 2023

Poznan University of Technology, Poznan, Poland

Thesis: A system for supporting the diagnosis of temporomandibular joint disorders

LANGUAGES

Polish (Native) | English (Full Professional Proficiency) | Indonesian (Intermediate)

INTERESTS

Linux DIY | Mnemonics | Language Learning | Generative AI Art | Strength Training & Systema